SLoW: Select Low-frequency Words! Automatic Dictionary Selection for Translation on Large Language Models

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Abstract

There are more than 7,000 languages around the world, and current Large Language Models (LLMs) only support hundreds of languages. Dictionary-based prompting methods can enhance translation on them, but most methods use all the available dictionaries, which could be expensive. Instead, it will be flexible to have a trade-off between token consumption and translation performance. This paper proposes a novel task called **A**utomatic **D**ictionary **S**election (**ADS**). The goal of the task is to automatically select which dictionary to use to enhance translation. We propose a novel and effective method which we call **S**elect **Lo**w-frequency **W**ords! (**SLoW**) which selects those dictionaries that have a lower frequency. Our methods have unique advantages. First, there is no need for access to the training data for frequency estimation (which is usually unavailable). Second, it inherits the advantage of dictionary-based methods, where no additional tuning is required on LLMs. Experimental results on 100 languages from FLORES indicate that SLoW surpasses strong baselines, and it can obviously save token usage, with many languages even surpassing the translation performance of the full dictionary baseline.[1][2]

1 Introduction

Large Language Models (LLMs) have exhibited many exciting capabilities such as chain-of-thought reasoning (Wang et al., 2023; Wei et al., 2024), neural machine translation (Lu et al., 2023; Zhu et al., 2024), code understanding and code generation (Li et al., 2023; Zhang et al., 2023), and even spatial reasoning (Hu et al., 2024). While LLMs have demonstrated their exciting performances on a wide range of tasks, they are usually English-centric, and their multilingual abilities are usually limited, especially in those low-resourced languages. Dictionary-based methods effectively improve multilingual capabilities by adding word mappings into the prompt (Lu et al., 2024; Lu et al., 2024). Yet, most current dictionary-based translation methods for LLMs use all the matching dictionaries greedily, and there are so far no systematic guidelines or architecture to select which dictionaries to use. Such a greedy strategy can lead to unnecessary token consumption, as LLMs may have no problems understanding some of the words. Furthermore, too much irrelevant or redundant information can distract LLMs (Shi et al., 2023). Therefore, we propose a novel Natural Language Processing task called **A**utomatic **D**ictionary **S**election (**ADS**). The input into the task of ADS is a set of available dictionaries and a set of input translation source instances. The goal of ADS is to maximise the translation performance by using only a subset of the dictionaries, so there can be a trade-off where more dictionaries to be used may have a better translation performance. Therefore, we constrain ADS to use no more than a certain number of words, $\mathcal{W}$ words, and in this paper, we make it the method with the lowest number of dictionaries among the methods in comparison.

To tackle ADS, we propose a novel and effective method which we call **S**elect **Lo**w-frequency **W**ords! (**SLoW**). SLoW selects the dictionaries that have a lower frequency in the training data. We postulate that this is because how frequently the words are presented in the training data is directly related to how well the LLMs understand them, and adding the dictionary of those low-frequency words makes it easier for LLMs to understand and translate less frequent and less well-learned words.

Further analysis indicates that such methods are better than many competitive baselines such as us-

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[1] A shocking fact is that there is no need to use the actual training data (often unobtainable) for frequency estimation, and an estimation frequency obtained using public resources is still apparently effective in improving translation with ChatGPT and LLaMa, and DeepSeek.

[2] https://github.com/HongyuanLuke/SLoW.

Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 898–913
November 4-9, 2025 ©2025 Association for Computational Linguistics

ing nouns, verbs, adjectives, or their combination greedily. Surprisingly, we also found that selecting partial dictionaries with SLoW can even beat full dictionary usage in some cases. This paves a new research direction to optimize the selection of dictionary usage automatically for dictionary-based translation methods on LLMs.

We emphasize the shocking fact that there is no need to obtain the actual training data, which is often unobtainable, and online public resources can be used to improve LLaMa and ChatGPT through SLoW. This suggests a good estimation of online resources on word frequencies in the training data from ChatGPT and LLaMa.

Our contributions are three-fold:

- We propose a novel task called **A**utomatic **D**ictionary **S**election, where it considers the trade-off between dictionaries and performance to be used when prompting LLMs.
- We propose a novel method to tackle ADS, which we call **S**elect **Lo**w-frequency **W**ords! (**SLoW**). SLoW selects the dictionaries that have a lower frequency in the training data.
- We conduct experiments on 100 languages from FLORES for Machine Translation. Experimental results indicate that SLoW beats competitive baselines and can even surpass the case when full dictionaries are used.

2 Prior Work

Neural Machine Translation via LLMs Research on effective methods for prompting English-centric Large Language Models (LLMs) for non-English tasks, including standard cross-lingual tasks like Multilingual Neural Machine Translation (MNMT), remains limited. Most existing studies have primarily focused on evaluating the translation performance of English-centric LLMs using prompts such as 'Translate to {language_name}: text' (Brown et al., 2020; Lin et al., 2022; Le Scao et al., 2022; Zhang et al., 2022). Various prompt formats have also been explored (Reynolds and McDonell, 2021; Wang et al., 2023). Additionally, Garcia and Firat (2022) examined the use of prompts to regulate aspects like formality or specific dialects in a generation. Furthermore, Agrawal et al. (2022) and Vilar et al. (2022) investigated selecting appropriate in-context examples to enhance the machine translation quality of LLMs. Generally speaking, the research of MNMT has now scaled to hundreds of languages as seen with FLORES (NLLB-Team, 2022).

Dictionary-based Method for Neural Machine Translation This research is closely tied to the concept of lexical constraints in machine translation, which can be categorized into hard constraints (Hokamp and Liu, 2017; Post and Vilar, 2018) and soft constraints (Song et al., 2019; Dinu et al., 2019; Chen et al., 2021).

Several studies have investigated the use of dictionaries in supervised machine translation. For instance, Zhang and Zong (2016) enhanced neural machine translation (NMT) by incorporating a bilingual dictionary to include rare or unseen words absent from the bilingual training data. Similarly, Arthur et al. (2016) improved the translation of rare words by integrating discrete translation lexicons and using the attention vector to estimate relevant lexical probabilities. Hämäläinen and Alnajjar (2020) leveraged dictionaries to generate synthetic parallel data, enhancing NMT training. Lu et al. (2024) used chained dictionaries to enhance machine translation with LLMs by leveraging intermediate auxiliary languages.

While much of the prior work has centred on using dictionaries for machine translation tasks, how to effectively select a subset of dictionaries to achieve a good trade-off remains unexplored. In contrast, ADS is the first task that considers which types of dictionaries should be used on LLMs for automatic machine translation.

3 Automatic Dictionary Selection

3.1 Translation with LLMs

We start by introducing our proposed task, namely automatic dictionary selection. The goal of such a task is to select appropriate dictionaries in order to maximise the performance of the succeeding generation task by adding the dictionaries into the prompt, and this paper focuses on the setting of neural machine translation on LLMs which use dictionaries for translation (Lu et al., 2024).

LLM can be regarded as a Seq2Seq neural network (Sutskever et al., 2014) to translate an input language into the output language while maintaining the semantical equivalence and maximise the following likelihood:

$$P\left(\hat{\boldsymbol{t}} \mid \boldsymbol{i}, \boldsymbol{s}, \boldsymbol{d}\right) = \prod_{j=1}^{\mathbb{T}} P\left(\hat{t}_j \mid \hat{t}_1, ..., \hat{t}_{j-1}, \boldsymbol{i}, \boldsymbol{s}, \boldsymbol{d}\right),$$

where $\mathbb{T}$ represents the length of the generated translation output and $\hat{t_j}$ represents the word at the position j that has been inferenced. $\boldsymbol{s}$ represents the source sentences, $\boldsymbol{d}$ represents the dictionaries that has been selected to be used for improving the translation. $\boldsymbol{i}$ represents the translation instruction to guide the LLMs to translate the words. A typical translation instruction could be:

> Translate the following sentence from <source language> into <target language>: <source sentence>

3.2 Automatic Dictionary Selection

However, which dictionaries to be used $\boldsymbol{d}$ has not been explored to our best knowledge. That means, in previous works, all the dictionaries are provided and inserted into the prompt as long as there is a match regardless of how useful they will be. However, intuitively speaking, this could not be the best choice. Even if the results are not maximised, one might want to reduce the computational cost as a trade-off to gain limited improvement with dictionary methods. Therefore, we propose a novel task ADS to automatically select dictionaries. The task is formulated as:

$$\hat{\mathcal{D}} = \mathcal{M}(\mathcal{D}, \mathcal{L}),$$

where $\mathcal{M}$ is a selection function, where we select a subset of dictionary $\hat{\mathcal{D}}$ from the complete dictionary $\mathcal{D}$ for succeeding downstream task dataset $\mathcal{L}$. Since such a selection might always be maximised by selecting the full dictionary, we define a dictionary size $\mathcal{V}$ which is usually lower than the full dictionary size, and the goal of ADS is to find a better function $\mathcal{M}$ that maximises the final performance on the $\mathcal{L}$ with a subset of the dictionary, namely, $\hat{\mathcal{D}}$, which has a dictionary size of $\mathcal{V}$.

3.3 Select Low-frequency Words!

In this paper, we propose a novel and effective method which we call **S**elect **Lo**w-frequency **W**ords! (**SLoW**). SLoW selects the dictionaries that have a lower frequency in the training data:

$$\hat{\mathcal{D}} = \text{first}(\text{sort}_{\bar{x}_i \in \mathcal{D}}(\mathcal{G}(\bar{x}_i, \mathcal{T})), \mathcal{V}), \quad (1)$$

where first selects from a sorted list in acending order created from sort to get the $\mathcal{V}$ lowest-frequency dictionaries selected by a frequency estimation function $\mathcal{G}$ with the training set $\mathcal{T}$ used for training the LLMs. Note that here for the translation task in this paper, English frequency can used as a standard, because most LLMs are English-centric.

We surprisingly found its usefulness despite it being simple, compared to various strong baselines that we have compared. This is yet intuitively aligned with our expectation, as we definitely would like to enhance LLMs' knowledge if that part of knowledge is not trained well. Data scarcity, i.e., low-frequency is a common reason for not training that part of the knowledge well.

In this paper, we have attempted various baselines. Since we conduct experiments on hundreds of languages from FLORES, and our computational resources are limited, we explore the setting where we set a fixed $\mathcal{V}$.

4 Experimental Setup

4.1 Datasets and Evaluation Metrics

We evaluate the task of Neural Machine Translation with the dictionary-based setting where dictionaries are used to improve machine translation (Lu et al., 2024). Under this setting, low-resourced languages play an important role, because dictionary-based methods are particularly useful on them (Lu et al., 2024). A very useful dataset is FLORES (NLLB-Team, 2022), where we use 100 languages from FLORES devtest. This dataset comprises 1,012 sentences sourced from English Wikipedia, spanning diverse topics and domains (we randomly sample 200 instances). These sentences have been meticulously translated into hundreds of languages by professional translators. Since they are professionally translated by human experts into parallel languages, it is suitable for our use.

For the evaluation metrics, we report the chrF (Popović, 2015) and the BLEU (Papineni et al., 2002) evaluations provided by the sacreBLEU repository.[3] We also use evaluate with COMET scores using wmt22-comet-da[4] (Rei et al., 2020) across all the experiments.

For space reasons, we present the language class of our experiments for XX translation in Table 9 and Table 10 in the Appendix.

4.2 Baselines

We conduct our experiments with both close-sourced and open-sourced LLMs on ChatGPT (GPT-4o-mini), LLaMa-3.1-8B (Dubey et al., 2024) and DeepSeek-V3 671B (DeepSeek-AI et al.,

[3] https://github.com/mjpost/sacrebleu
[4] https://github.com/Unbabel/COMET

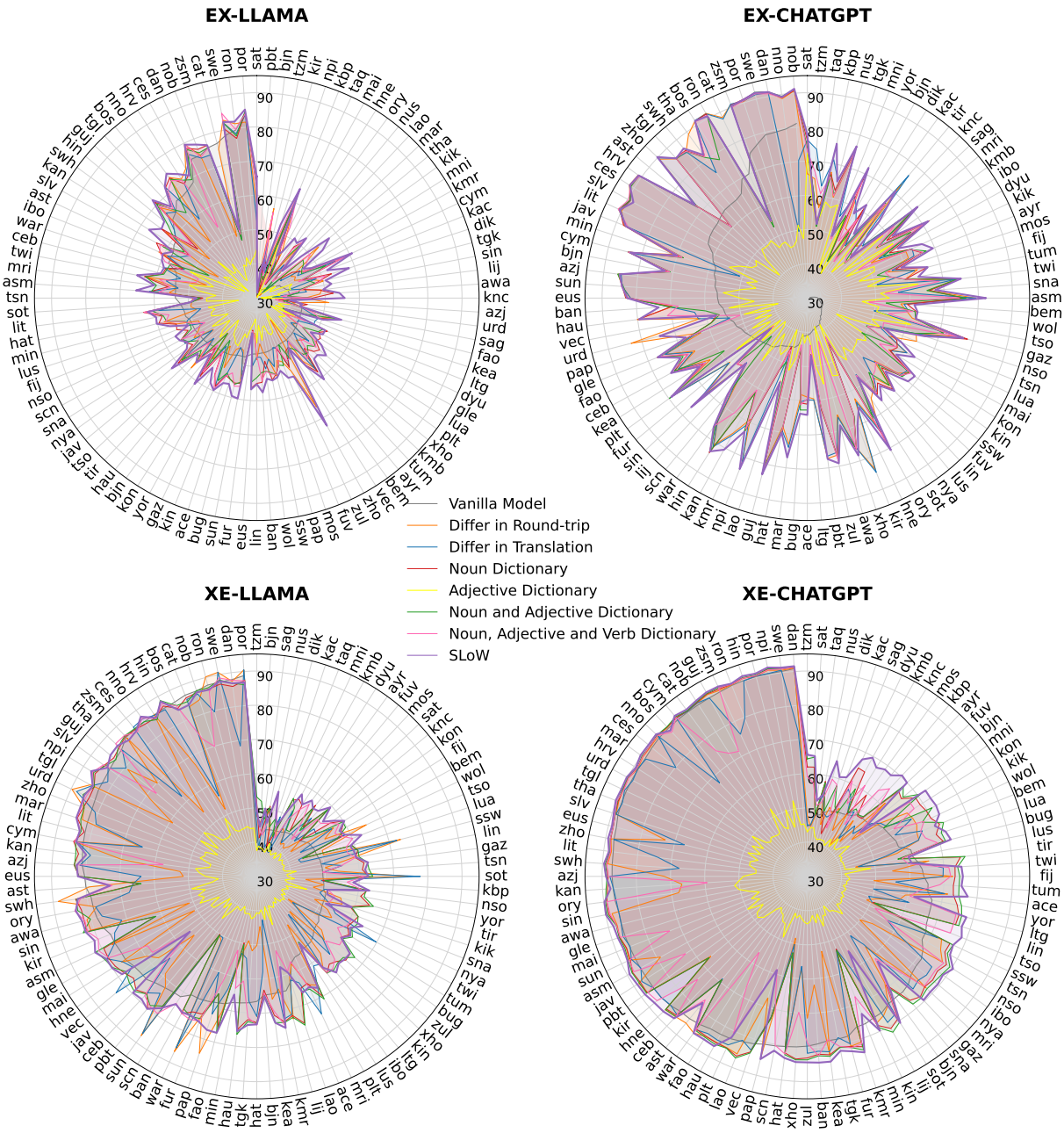


Figure 1: Performance of LLaMa and ChatGPT in COMET scores on the task of Machine Translation both into English and from English translation on FLORES with different ADS methods. The top-left one is the translation from English on LLaMa, the top-right is the translation from English on ChatGPT, the bottom-left is the translation to English on LLaMa, and the bottom-right is the translation to English on LLaMa. It is obvious that our proposed method SLoW is the best, surpassing many strong baselines. Such a phenomenon can be consistently observed across many low-resourced and high-resourced languages, demonstrating the effectiveness of our methods. For space reasons, more results on BLEU, chrF, evaluations and on DeepSeek-V3 in the Appendix in Table 4.

2024). At the time of writing, both of them are popular and widely used English-centric LLMs which are strong in their multilingual translation capacities. Based on these popular LLMs, we compare our proposed method to strong baseline methods:

- **Vanilla Model** We prompt LLMs to directly translate the input without the assistance of any additional dictionaries.
- **Noun Dictionary** Noun words may contain named entities which can be special terminologies which could be particularly hard to translate (Ugawa et al., 2018).
- **Adjective Dictionary** Adjective words are another type of word which could be important.

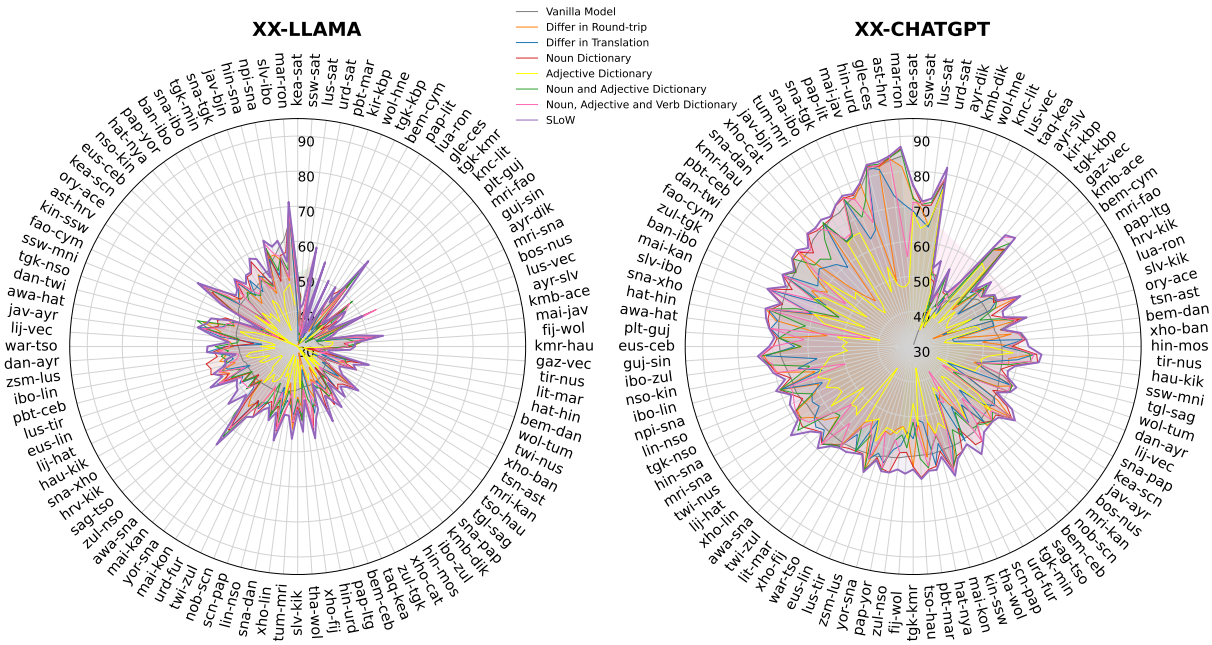


Figure 2: Performance of LLaMa and ChatGPT the task of Machine Translation on non-English-centric translation in COMET scores on non-English-centric translation on FLORES with different ADS methods. It is obvious that our proposed method SLoW is the best, surpassing many strong baselines. Such a phenomenon can be consistently observed across many translation pairs, demonstrating the effectiveness of our methods. For space reasons, more results on BLEU, chrF, evaluations and on DeepSeek-V3 in the Appendix in Table 4.

Direction	# improved > 5 points > 10 points > 20 points				# degraded > 5 points > 20 points		
En-X-LLaMa	88/100	65/88	50/88	22/88	12/100	4/12	3/12
En-X-CHATGPT	75/100	36/75	15/75	1/75	25/100	10/25	4/25
X-En-LLaMa	76/100	63/76	57/76	39/76	24/100	10/24	5/24
X-En-CHATGPT	92/100	50/92	43/92	33/92	8/100	5/8	2/8
X-X-LLaMa	93/100	46/93	7/93	1/93	7/100	1/7	0/7
X-X-CHATGPT	100/100	53/100	26/100	5/100	0/100	0/0	0/0

Table 1: Statistics of the changes in COMET scores with SLoW compared to the baseline of Differ in Round-trip on LLaMa and ChatGPT on the FLORES dataset. Most translation directions have been obviously improved. Details results on BLEU, chrF, and evaluations on DeepSeek-V3 can be found in Table 4.

- **Verb Dictionary** Verb words are another type of word which could be important.
- **Noun and Adjective Dictionary** Combining both noun and adjective dictionaries can be useful as well.
- **Noun, Adjective, and Verb Dictionary** Combining noun, adjective, and verb dictionaries can be useful as well.
- **Differ in Round-trip** We first use a baseline model without any dictionary to translate the source language into the target language before translating back (Sennrich et al., 2016). The difference between the round-trip translation and the original source sentences is selected as the dictionary.
- **Differ in Translation** We first use a baseline without any dictionary to translate the source language into the target language. The difference between the translation and the original target sentences is selected as the dictionary.

4.3 Frequency Estimation

Since the training sets of LLMs are usually close-sourced, we estimate the word frequency of training data by directly using web resources.[5]

[5] https://github.com/rspeer/wordfreq

4.4 Prompt Template

Dictionary Construction To construct the bilingual dictionary mapping for translation, we prompt ChatGPT (Lu et al., 2024):

> (1) Please provide the translation of the given English sentence into <language>, along with a word-for-word dictionary for each word.
> (2) The output format must be strictly followed:
> 1. Start with 'English:' followed by the English sentence.
> 2. On the next line, start with '<language>:' followed by the <source> translation.
> 3. On the next line, start with 'dictionary:' followed by each word in the <language> sentence, annotated with its English meaning in parentheses, separated by spaces.
> (3) Now generate translations for the following sentence:
> English: <target>
> <language>: <source>
> dictionary:

Translation We leave the translation prompt in the Appendix due to space reasons.

5 Results

5.1 Main Results

From-English Translation (EX) The upper part in Figure 1 visually demonstrates the performance of SLoW on the task of Machine Translation with ADS on the dataset of FLORES compared to strong baselines. The top-left figure shows the performance of LLaMa from English to other languages. The average performance seems to be the lowest among all four figures, which is reasonable. One reason is that this is the translation from English, which is usually lower than translating into English on the English-centric model on average. Another reason is that it is usual for an 8B version LLaMa to be less powerful than close-sourced ChatGPT 4.

The top-right figure shows the translation performance of ChatGPT from English to other languages. It is obvious that the average performance is better than the from-English direction on LLaMa. This is also reasonable that it is slightly better than the bottom-left figure on translation to English on LLaMa, as to English translation can be considered as generally better than from-English translation on the English-centric model.

Overall, SLoW (purple line) performs clearly better than all the other baselines when translating from English. On LLaMa, it seems that the advantage of SLoW is more clear than on ChatGPT compared to Differ in Round-trip. One postulation is that the performance of LLaMa is generally lower than ChatGPT, so there is more room for improvement for SLoW. Generally speaking, SLoW is clearly useful in improving both LLaMa and ChatGPT on translating from English.

Table 1 presents the improvement statistics of SLoW compared to the baseline of Differ in Round-trip for translating from English. SLoW surpasses the baseline. For example, 88 out of 100 language pairs are improved when using SLoW for translating from English for LLaMa. Among those 88 pairs, 22 (25%) of the pairs are improved for more than 20 COMET scores. In comparison, the number of degradations is apparently lower (12 out of 100 language pairs). When there is a degradation, about half of the language pairs (5/12) give less than 5 points of degradation. These results highlight the usefulness of SLoW.

Into-English Translation (XE) The lower part in Figure 1 visually demonstrates the performance of SLoW on the task of Machine Translation with ADS on the dataset of FLORES compared to strong baselines. The bottom-left figure shows the translation performance of LLaMa from English to other languages. The average performance seems to be lower than translating to English on ChatGPT, but higher than translating from English on LLaMa, which is reasonable. One reason is that this is the translation from English, which is usually lower than translating into English on the English-centric model on average. Another reason is that it is usual for an 8B version LLaMa to be less powerful than close-sourced ChatGPT 4.

The bottom-right figure shows the translation performance of ChatGPT into English. It is obvious that the average performance is better than the performance in all the other three figures. This is because translating into English is usually easier than translating from English, and ChatGPT-4 can be usually considered than LLaMa-3.1-8B.

Overall, SLoW (purple line) performs clearly better than all the other baselines for translating from English translation. On LLaMa, it seems that the advantage of SLoW is more clear than on Chat-

PoS	Tag	Per.	Cov.	Per.	Cov.	Per.	Cov.
		XE		*EX*		*XX*	
adjective	ADJ	19.28%	66.82%	19.30%	58.29%	19.16%	62.88%
adposition	ADP	2.26%	6.10%	2.34%	6.36%	2.51%	7.74%
adverb	ADV	4.33%	41.73%	4.13%	34.92%	4.53%	42.93%
auxiliary	AUX	0%	0%	0%	0%	0%	0%
coordinating conjunction	CCONJ	0.34%	4.01%	0.15%	1.52%	0.20%	2.20%
determiner	DET	0.45%	1.40%	0.44%	2.41%	0.58%	3.53%
interjection	INTJ	0%	0%	0%	0%	0%	0%
noun	NOUN	43.65%	49.23%	46.45%	45.74%	45.61%	50.80%
numeral	NUM	3.97%	53.51%	3.35%	42.95%	3.34%	48.10%
particle	PART	0.11%	14.68%	0.10%	26.32%	0.11%	32.82%
pronoun	PRON	0.68%	7.82%	0.53%	7.03%	0.14%	90.12%
proper noun	PROPN	0.14%	92.15%	0.12%	93.83%	0.14%	90.12%
punctuation	PUNCT	0%	0%	0%	0%	0%	0%
subordinating conjunction	SCONJ	0%	0%	0%	0%	0%	0%
symbol	SYM	0%	0%	0%	0%	0%	0%
verb	V	24.06%	46.27%	22.58%	41.85%	22.58%	47.00%
others	X	0.73%	6.71%	0.50%	5.85%	0.66%	8.04%

Table 2: PoS tagger statistics selected by SLoW. Per. represents the percentage of the tag in the whole dictionary prompted, and Cov. represents the coverage, meaning the selected ratio of the selected words compared to the total number of that PoS tag in the dictionary. There are 17 core tags in UPoS: https://universaldependencies.org/u/pos/ UPoS tagger: https://github.com/slavpetrov/universal-pos-tags, https://www.nltk.org/.

GPT compared to Differ in Round-trip. One postulation is that the performance of LLaMa is generally lower than ChatGPT, so there is more room for improvement for SLoW. Generally speaking, SLoW is clearly useful in improving both LLaMa and ChatGPT on translating from English.

Table 1 presents the improvement statistics of SLoW compared to the baseline of Differ in Round-trip for translating into English. It is obvious that SLoW surpasses the baseline. For example, when translating into English on ChatGPT, 92 out of 100 language pairs are improved when using SLoW. Among those 92 pairs, 33 (about 1/3) of the pairs are improved for more than 20 COMET scores. In comparison, the number of degradations is lower (8 out of 100 language pairs). When there is a degradation, about half of the language pairs (5/8) give less than 20 points of degradation. This highlights the usefulness of SLoW.

Non-English-centric Translation (XX) Figure 2 visually demonstrates the performance of SLoW on the task of Machine Translation with ADS on the dataset of FLORES compared to strong baselines. The left figure is for translation on LLaMa and the right figure is for translation on ChatGPT. The overall performance on LLaMa is apparently lower than ChatGPT, which is reasonable, as the model size of LLaMa we conduct our experimentation on is obviously smaller than ChatGPT.

Overall, it is clear that SLoW is great at translating in non-English-centric directions, surpassing all the strong baselines. We also note that the baselines might not work well in this scenario, as they can be frequently worse than the Vanilla Baseline, which does not use any additional dictionary. We postulate that this is due to the lack of ability in terms of non-English-centric translation, as ChatGPT is an English-centric model. Still, SLoW is effective and constantly better than the Vanilla Baseline.

Table 1 presents the improvement statistics of SLoW compared to the baseline of Differ in Round-trip for translating on non-English-centric directions. It is obvious that SLoW surpasses the baseline. For example, when translating into English on ChatGPT, 100 out of 100 language pairs are improved when using SLoW. Among those 92 pairs, 46 (about half) of the pairs are improved for more than 5 COMET scores. In comparison, the number of degradations is lower, for example, on LLaMa, only 0 out of 100 language pairs are degraded. This highlights the usefulness of SLoW.

Ground-truth Tsonga	Muteki wa swifaniso u tleketliwile ku ya le Ronald Reagan UCLA Medical Center, laha a fikeke a fa.
Ground-truth English	The photographer was transported to Ronald Reagan UCLA Medical Center, where he subsequently died.
Vanilla Model	COMET Score: 0.579 ✗ "Ambulance from Ronald Reagan UCLA Medical Center has arrived."
Differ in Round-trip	COMET Score: 0.413 ✗ The ensued entertainment after Ronald Reagan UCLA Medical Centre called the police but attempted him for throad and for traffic stop due to him driving too fast therefore prompting visual imagery for the first time in South Africa.
Differ in Translation	COMET Score: 0.335 ✗ The driver of the police vehicle had his ear given according to the witness who is accused of not being prepared to pay a fine.
Noun Dictionary	COMET Score: 0.806 ✗ The person in the photographs was taken to the Ronald Reagan UCLA Medical Center, where he died.
Adjective Dictionary	COMET Score: 0.562 ✗ Ronald Reagan Center where he died.
Noun and Adjective Dictionary	COMET Score: 0.763 ✗ The driver of the vehicle was taken to Ronald Reagan UCLA Medical Center, where he died.
Noun, Adjective, and Verb Dictionary	COMET Score: 0.864 ✗ The photographer was taken to the Ronald Reagan UCLA Medical Center, where he died.
High-frequency	COMET Score: 0.707 ✗ The man of images was taken to the Ronald Reagan UCLA Medical Center, where he died.
SLoW	COMET Score: 0.912 ✓ The photographer was transported to the Ronald Reagan UCLA Medical Center, where he subsequently died.

Table 3: A case study on translating from Tsonga To English. ✗ represents that the generation is not the best among all the models. ✓ represents that the generation is the best among all the models.

5.2 SLoW PoS Tags

Table 2 presents the PoS tags of the words selected by SLoW. The dictionary is mainly composed of adjective, noun, and verb words. SLoW surpasses the baseline, which is composed of only these three types of words without considering how frequent they are. In contrast, SLoW selects low-frequency words appropriately, such as numerals and adverbs. However, it could be expensive to run exhaustive experiments on all combinations to be compared with SLoW. Nevertheless, the statistics suggest that SLoW selects a comprehensive dictionary composed of diverse words with different PoS tags, which is effective in improving the translation. This also surpasses the strong baseline with Differ in Round-trip and Differ in Translation.

We present further results on BLEU, chrF evaluations, and results on DeepSeek-V3 in Table 4. We also leave case studies in the Table 3. They all align with our conclusions. On most language pairs, the performance has been obviously improved. In case there is any degradation, the degradation is frequently less than 1 point. For space reasons, we leave more case studies in our Appendix.

5.3 SLoW versus Full Dictionary

While usually adding redundant information to LLMs can degrade performance, removing useful dictionaries can be harmful to translation performance. Table 5 presents the actual ratio that we have adopted compared to the full dictionary. We also note that under this setting, SLoW can surpass the full dictionary baseline obviously on some language pairs as presented in Table 6. Yet, for most other cases, the full dictionary is still better, which is however still reasonable and very acceptable as more tokens are cost with LLMs. We also note that there is still a chance for SLoW to surpass the full dictionary baseline better if a different ratio is

Direction	# improved	> 1 point	> 2 points	> 3 points	> 5 points	# degraded	> 1 point	> 2 points	> 3 points	> 5 points
EX-CHATGPT-BLEU	72/100	48/72	30/72	20/72	11/72	28/100	9/28	2/28	0/28	0/28
EX-CHATGPT-chrF	69/100	52/69	30/69	20/69	12/69	31/100	13/31	2/31	0/31	0/31
XE-CHATGPT-BLEU	70/100	50/70	26/70	14/70	11/70	30/100	14/30	8/30	7/30	7/30
XE-CHATGPT-chrF	70/100	41/70	25/70	16/70	13/70	30/100	16/30	9/30	7/30	7/30
XX-CHATGPT-BLEU	69/100	37/69	19/69	6/69	1/69	31/100	6/31	0/31	0/31	0/31
XX-CHATGPT-chrF	69/100	42/69	19/69	8/69	1/69	31/100	11/31	0/31	0/31	0/31
EX-LLaMa-BLEU	85/100	73/85	59/85	52/85	30/85	15/100	8/15	5/15	5/157	3/15
EX-LLaMa-chrF	71/100	50/71	32/71	20/71	13/71	29/100	13/29	5/29	2/29	0/29
XE-LLaMa-BLEU	58/100	37/58	17/58	12/58	10/58	42/100	28/42	11/28	8/28	5/28
XE-LLaMa-chrF	64/100	40/64	24/64	18/64	12/64	36/100	24/36	15/36	7/36	6/36
XX-LLaMa-BLEU	84/100	54/84	39/84	28/84	6/84	16/100	6/16	2/16	2/16	0/16
XX-LLaMa-chrF	80/100	70/80	50/80	29/80	11/80	20/100	10/20	7/20	5/20	0/10
EX-DEEPSEEKV3-BLEU	68/100	47/68	25/68	17/68	12/68	32/100	12/32	2/32	0/32	0/32
EX-DEEPSEEKV3-chrF	73/100	48/73	31/73	21/73	14/73	27/100	13/27	4/27	1/27	0/27
XE-DEEPSEEKV3-BLEU	66/100	44/66	30/66	17/66	13/66	34/100	19/34	9/34	8/34	7/34
XE-DEEPSEEKV3-chrF	74/100	53/74	35/74	19/74	12/74	26/100	11/26	9/26	9/26	7/26
XX-DEEPSEEKV3-BLEU	72/100	38/72	13/72	4/72	0/72	28/100	4/28	0/28	0/28	0/28
XX-DEEPSEEKV3-chrF	75/100	48/75	25/75	8/75	2/75	25/100	8/25	1/25	0/25	0/25

Table 4: Statistics of the changes in BLEU and chrF scores with SLoW compared to the baseline of the Noun Dictionary on CHATGPT, LLaMa, and DEEPSEEK-V3. Most translation directions have been obviously improved. In case there is any degradation, the degradation is frequently less than 1 point.

Direction	Ratio
into-English	0.553
from-English	0.520
non-English-centric	0.564

Table 5: The dictionary ratio is automatically decided in this paper by aligning with the word numbers in Differ in Round-trip compared to the full dictionary.

Direction	SLoW	Full-Dict
pbt_Arab	0.803	0.483
kir_Cyrl	0.810	0.501
gle_Latn	0.802	0.499
ory_Orya	0.839	0.518
azj_Latn	0.827	0.514

Table 6: Five XE translation pairs on LLaMa, showing that SLoW obviously surpasses the Full-Dict baseline.

chosen. Since this is too exhaustive for this paper, we leave the exploration to future work.[6]

5.4 SLoW versus High-frequency

In order to further validate our claim that lower-frequency dictionaries are more useful for translation than higher-frequency ones, we perform a comparison between SLoW and those dictionaries with the highest frequency and present the results in Table 7. When the same number of words and PoS ratios are kept, we see that SLoW is clearly better in high-frequency dictionaries. For example, for translating into English on ChatGPT, SLoW

[6]For baselines with more/fewer words than this ratio, random padding or dropping is adopted.

Direction	High-frequency	SLoW
XE-ChatGPT	0	100
EX-ChatGPT	8	92
XX-ChatGPT	16	84
XE-LLaMa	19	81
EX-LLaMa	1	99
XX-LLaMa	13	87

Table 7: The number of winning languages in COMET scores on different language pairs and different models with High-frequency dictionaries and SLoW.

is always better than high-frequency dictionaries. This apparently strengthens the claim of this paper.

6 Conclusions

LLMs are highly effective in English but underperform in many other languages, especially low-resourced ones. Using dictionary-based methods can improve translation performance, but previous research has not investigated which dictionaries can be more useful to LLMs and they usually add all the dictionaries to the prompt. To this end, we propose a novel method called **S**elect **Lo**w-frequency **W**ords! (**SLoW**). Given the number of dictionaries to be selected, SLoW selects those with the lowest frequency. We found that such a novel and effective algorithm achieves strong performance, clearly surpassing many strong baselines, including high-frequency dictionaries. Also, general web resources can be used to estimate the frequency instead of the actual training data of the LLMs.

Limitations

This paper presents an analysis of 100 languages only. However, there are more than 7,000 languages around the world. The paper can be further extended by including more languages; however, such datasets are lacking. It is quite valuable as the data collection procedure itself is hard, which can significantly contribute to our community.

Second, since we have no access to the training data of the LLMs that we conduct experiments on, it is a pity that we cannot use them to estimate the actual word frequency for experimental purposes.

Ethical Statement

We honour and support the ACL ARR Code of Ethics. There is no ethical issue known to us. Well-known and widely used LLMs are used in our work, which is subjected to generating offensive context. Yet, the above-mentioned issues are widely known to exist commonly among LLMs. Any content generated does not reflect the view of the authors.

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Appendix

Ground-truth English	The Iraq Study Group presented its report at 12.00 GMT today.
Ground-truth Standard Malay	Kumpulan Kajian Iraq telah membentangkan laporan mereka pada pukul 12.00 GMT hari ini.
Vanilla Model	COMET Score: 0.944 ✗ Kumpulan Kajian Iraq melepaskan laporan mereka pada pukul 12.00 GMT hari ini.hobbies
Back to English	The Iraq Study Group released their report at 12:00 GMT today.
Differ in Round-trip	COMET Score: 0.941 ✗ Kumpulan Kajian Iraq melaporkan laporan mereka pada pukul 12.00 GMT hari ini.
Back to English	The Iraq Study Group reported their report at 12:00 GMT today.
Differ in Translation	COMET Score: 0.915 ✗ Kajian mengenai Iraq telah membentangkan laporan mereka pada jam 1200 pagi ini.
Back to English	The study on Iraq has presented their report at 12:00 this morning.
Noun Dictionary	COMET Score: 0.949 ✗ Kumpulan Kajian Iraq memperkenalkan laporan mereka pada jam 12.00 GMT hari ini.
Back to English	The Iraq Study Group introduced their report at 12:00 GMT today.
Adjective Dictionary	COMET Score: 0.910 ✗ Kumpulan Kaji Iraq memperkenalkan laporannya pada jam 12.00 GMT pagi ini.
Back to English	The Iraq Study Group introduced its report at 12:00 AM GMT this morning.
Noun and Adjective Dictionary	COMET Score: 0.946 ✗ Kumpulan Kajian Iraq memperkenalkan laporannya pada pukul 12.00 GMT hari ini.
Back to English	The Iraq Study Group introduced its report at 12.00 GMT today.
Noun, Adjective, and Verb Dictionary	COMET Score: 0.894 ✗ Kumpulan Kaji Iraq mengemukakan laporan mereka pada masa GMT pukul 12 pagi hari ini.
Back to English	The Iraq Study Group submitted their report at 12 AM GMT today.
High-frequency	COMET Score: 0.940 ✗ Kumpulan Kajian Iraq memperkenalkan laporannya ini pada jam 12.00 GMT hari ini.
Back to English	The Iraq Study Group introduced its report at 12.00 GMT today.
SLoW	COMET Score: 0.976 ✓ Kumpulan Kajian Iraq telah membentangkan laporan mereka pada pukul 12.00 GMT hari ini.
Back to English	The Iraq Study Group has presented their report at 12:00 GMT today.

Table 8: A case study on translating from English to Standard Malay. ✗ represents that the generation is not the best among all the models. ✓ represents that the generation is the best among all the models.

Language Class	Number
0	19
1	41
2	9
3	10
4	5
Total	84

Table 9: A list of language classes of the 100 languages used in our experiments. More than half of the languages used in our study are relatively low-resource according to Joshi et al. (2020).

Language Pairs	Number
0 —> 0	4
0 —> 1	9
0 —> 2	2
0 —> 3	4
0 —> 4	1
1 —> 0	6
1 —> 1	31
1 —> 2	7
1 —> 3	4
1 —> 4	1
2 —> 0	3
2 —> 1	8
2 —> 2	0
2 —> 3	1
2 —> 4	2
3 —> 0	2
3 —> 1	7
3 —> 2	2
3 —> 3	0
3 —> 4	0
4 —> 0	1
4 —> 1	3
4 —> 2	0
4 —> 3	2
4 —> 4	0
Total	100

Table 10: A list of language pair classes of the XX translation experiments. More than half of the languages used in our study are relatively low-resource according to Joshi et al. (2020).

We use the following prompt for translation:

> Translate the following sentence from {source_language} to {target_language}.
> {origin_sentence}
> Use the provided dictionary to clarify or improve the translation of any misaligned words.
> - Here are some dictionaries that you need to focus on:
> {dict}
> Note: Finally, only respond to me with the final {target_language} translation. Your output format is as follows:
> The refined translation is:

The dictionary size for the constructed dictionary is: EX: 1581.62, XE: 1539.72, XX: 1636.59, averaged from all languages in our experiments. For each prompt, the number of dictionaries to be used in all models is aligned with the baseline Differ in Round-trip throughout experiments.